Example: DSP algorithms for controlling echoes in long distance telephone calls.

Reconstruction filter: A low-pass analog filter placed after a digital-to-analog converter. Smoothes the stepped waveform by removing frequencies above one-half the sampling rate.

Rectangular form: Representing a sinusoid by the form: $A\cos(\omega t) + B\sin(\omega t)$, where A is called the *real part* and B is called the *imaginary part* (even though these are not imaginary numbers).

Rectangular window: A signal with a group of adjacent points having unity value, and zero elsewhere. Usually multiplied by another signal to select a section of the signal to be processed.

Recursion coefficients: The weighing values used in a recursion equation. The recursion coefficients determine the characteristics of a recursive (IIR) filter.

Recursion equation: Equation relating the past and present samples of the output signal with the past and present values of the input signal. Also called a *difference equation*.

Region-of-convergence: The term used in the Laplace and z-transforms. Those regions in the s-plane and z-planes that have a defined value.

RGB encoding: Representing a color image by specifying the amount of red, green, and blue for each pixel.

RISC: Reduced Instruction Set Computer, also called a DSP microprocessor. A fewer number of programming commands allows much higher speed math calculations. The opposite is the Complex Instruction Set Computer, such as the Pentium.

ROC curve: A graphical display showing how threshold selection affects the performance of a target detection problem.

Roll-off: Jargon used to describe the sharpness of the transition between a filter's passband and stopband. A *fast* roll-off means the transition is sharp; a *slow* roll-off means it is gradual.

Root-mean-square (rms): Used to express the fluctuation of a signal around *zero*. Often used in electronics. Defined as the square-root of the mean of the squares. See *standard deviation* for comparison.

Round-off noise: The error caused by rounding the result of a math calculation to the nearest quantization level.

Row major order: A pattern for converting an image to serial form. Operates the same as English writing: left-to-right on the first line, left-to-right on the second line, etc.

Run-length encoding: Simple data compression technique with many variations. Characters that are repeated many times in succession are replaced by codes indicating the character and the length of the run.

Running sum: An operation used with discrete signals that mimics integration of continuous signals. Also called the *discrete integral*.

s-domain: The domain defined by the Laplace transform. Also called the *s-plane*.

Sample spacing: The spacing between samples when a continuous image is digitized. Defined as the center-to-center distance between pixels.

Sampling aperture: The region in a continuous image that contributes to an individual pixel during digitization. Generally about the same size as the sample spacing.

Sampling theorem: If a continuous signal composed of frequencies less than f is sampled at $2f$, all of the information contained in the continuous signal will be present in the sampled signal. Frequently called the *Shannon* sampling theorem or the *Nyquist* sampling theorem.

SECAM: Television standard used in Europe. See NTSC for comparison.

Seismology: Branch of geophysics dealing with the mechanical properties of the earth.

Separable: An image that can be represented as the product of its vertical and horizontal profiles. Used to improve the speed of image convolution.

Sharpening: Image processing operation that makes edges more abrupt.

Shift and subtract: Image processing operation that creates a 3D or embossed effect.

Shift invariance: A property of many systems. A shift in the input signal produces nothing more than a shift in the output signal. Means that the characteristics of the system do not changing with time (or other independent variable).